

2. Sodium oxide is formed when sodium burns in oxygen.

(a) Use the Periodic Table on page 20 to describe the structure of the oxygen atom. [3]

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(b) During the production of sodium oxide the Na^+ and O^{2-} ions are formed.

(i) Give the electronic structures of the ions. [2]

Na^+

O^{2-}

(ii) Sodium reacts violently with water to form a strong alkali.
State the name of the alkali formed in this reaction. [1]

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(c) Small amounts of sodium peroxide, Na_2O_2 , are also formed when sodium burns.
Calculate the relative formula mass (M_r) of sodium peroxide. [2]

A_r : Na = 23, A_r : O = 16

M_r =

